Blessed G/Michael Catholic Secondary School

Semester 1 Chemistry Mid1 Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** For a reaction with rate law Rate = k[A]^2[B], what is the overall order of the reaction?

A. First order **B.** Second order **C.** Third order **D.** Zero order

____ **2.** What are the SI units of the rate constant (k) for a first-order chemical reaction?

A. mol / (L * s) **B.** s^(-1)

C. L / (mol * s) **D.** L^2 / (mol^2 * s)

____ **3.** How does increasing the temperature affect the rate constant (k) according to the Arrhenius equation?

A. Increases k exponentially **B.** Decreases k

C. Has no effect on k **D.** Decreases activation energy Ea

____ **4.** A catalyst speeds up a chemical reaction by:

A. Increasing enthalpy of reaction Delta H
B. Providing an alternative pathway with lower activation energy
C. Increasing the frequency of non-effective collisions
D. Shifting the chemical equilibrium position to the right

____ **5.** The half-life of a first-order reaction is:

A. Independent of initial reactant concentration
B. Inversely proportional to initial concentration
C. Directly proportional to initial concentration
D. Zero at equilibrium

____ **6.** In an exothermic reversible reaction, what happens to the equilibrium constant Kc when the temperature is raised?

A. Kc increases **B.** Kc decreases

C. Kc remains unchanged **D.** Kc drops to zero

____ **7.** What is the pH of a 0.001 M hydrochloric acid (HCl) solution at 25°C?

A. 1 **B.** 2 **C.** 3 **D.** 11

____ **8.** In the reaction 2NO + O2 -> 2NO2, doubling [NO] while keeping [O2] constant quadruples the rate. What is the order with respect to NO?

A. Second order **B.** First order **C.** Zero order **D.** Third order

____ **9.** What is the relationship between Kp and Kc for the gaseous equilibrium: N2(g) + 3H2(g) <=> 2NH3(g)?

A. Kp = Kc * (RT)^(-2) **B.** Kp = Kc * (RT)^2

C. Kp = Kc * (RT) **D.** Kp = Kc

____ **10.** Which of the following plots yields a straight line for a second-order reaction (A -> Products)?

A. 1 / [A] versus time t **B.** ln[A] versus time t

C. [A] versus time t **D.** sqrt([A]) versus time t

____ **11.** According to collision theory, for a chemical collision to be effective, molecules must have:

A. Both energy equal to or greater than activation energy (Ea) and correct spatial orientation
B. Sufficient kinetic energy only
C. Correct orientation only
D. Low molecular mass

____ **12.** For the Haber process N2(g) + 3H2(g) <=> 2NH3(g) (Delta H = -92 kJ/mol), increasing pressure shifts equilibrium:

A. Toward products (to the right) **B.** Toward reactants (to the left)

C. Has no effect on equilibrium **D.** Stops reaction completely

____ **13.** What is the conjugate base of the bisulfate ion (HSO4-)?

A. Sulfate ion (SO4^2-) **B.** Sulfuric acid (H2SO4)

C. Sulfur trioxide (SO3) **D.** Hydronium ion (H3O+)

____ **14.** A solution with pH = 4 has a hydrogen ion concentration [H+] that is how many times greater than a solution with pH = 6?

A. 100 times **B.** 2 times **C.** 10 times **D.** 1000 times

____ **15.** What is the ionic product constant of pure water (Kw) at 25°C?

A. 1.0 * 10^(-14) **B.** 1.0 * 10^(-7) **C.** 1.0 * 10^(-1) **D.** 7.0

____ **16.** A buffer solution resists changes in pH when small amounts of acid or base are added. It is typically prepared from:

A. A weak acid and its conjugate base (salt)
B. A strong acid and a strong base
C. Pure sodium chloride in water
D. A concentrated strong acid only

____ **17.** What is the Henderson-Hasselbalch equation used to calculate buffer pH?

A. pH = pKa + log([Base] / [Acid]) **B.** pH = pKa - log([Base] / [Acid])

C. pH = -log(pKa) **D.** pH = pKa * [Base] / [Acid]

____ **18.** If the reaction quotient Q is greater than the equilibrium constant Kc (Q > Kc), the reaction will:

A. Proceed in the reverse direction (to the left) toward reactants
B. Proceed in the forward direction
C. Remain at equilibrium
D. Precipitate completely

____ **19.** The rate-determining step in a multi-step reaction mechanism is:

A. The slowest elementary step **B.** The fastest elementary step

C. The first step always **D.** The step with lowest activation energy

____ **20.** What is the solubility product constant (Ksp) expression for silver chloride AgCl(s) <=> Ag+(aq) + Cl-(aq)?

A. Ksp = [Ag+][Cl-] **B.** Ksp = [Ag+][Cl-] / [AgCl]

C. Ksp = [Ag+] + [Cl-] **D.** Ksp = [Ag+]^2

____ **21.** The common-ion effect describes the phenomenon where the solubility of a sparingly soluble salt:

A. Decreases when a common ion is introduced into the solution
B. Increases dramatically
C. Remains unchanged
D. Becomes infinite

____ **22.** Which of the following is a Lewis acid?

A. BF3 (Boron trifluoride) **B.** NH3 (Ammonia)

C. OH- (Hydroxide) **D.** H2O

____ **23.** In the Arrhenius equation k = A * e^(-Ea / RT), the pre-exponential factor A is related to:

A. Collision frequency and steric orientation factor
B. Enthalpy change
C. Universal gas constant
D. Reaction order

____ **24.** What is the pOH of a solution with [OH-] = 1.0 * 10^(-5) M?

A. 5 **B.** 9 **C.** 14 **D.** 7

____ **25.** For a zero-order reaction A -> Products, the rate of reaction is:

A. Constant and independent of reactant concentration
B. Directly proportional to [A]
C. Proportional to [A]^2
D. Exponentially declining